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System
Data
Drive
Vendor

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Lecturer
CSE.

- Defⁿ of f^n
- Domain and Range of f^n
- Limits of f^n
- Different types of f^n
- Graph of different type of f^n
- Continuity and differentiability

1. PK. Bhattacharjee → Differential Calculus.
2. Abu Yusuf → "
3. Das & Mukherjee → "
4. Frank Ayres (Schaum's outline series) → Calculus.
5. H.S.C. text books.
6. S.S.C. → Higher Maths.
7. Google & Khan Academy.

[illegible]

Chapter - 1

Functions

* Definition of function:

A function is a rule that associates with each value of a variable x in a certain set, exactly one value of another variable y . The variable y is called the dependent variable, and x is called the independent variable.

Symbolically, we write, $y = f(x)$ or $y = \phi(x)$.

For example, $y = x^2$ or $y = 3x + 5$ or $A = \mathbb{R}^n$

Note:- Function as a relation: Let X and Y be two non empty sets. Then a relation from X to Y with domain X is called a function, if no two different ordered pairs in f have the same first co-ordinates.

(i) Example:- Let, $X = \{0, 1, 4, 9\}$ and $Y = \{0, 1, 2, -2, 3\}$

Let, $f = \{(0, 0), (1, 1), (4, 2), (4, -2), (9, 3)\}$

Clearly two different ordered pairs namely $(4, 2)$ and $(4, -2)$ have the same first co-ordinates and therefore f is a relation but not a function.

(ii) Let, $X = \{1, -1, 2, -2\}$ and $Y = \{1, 4, 5\}$.

and let, $g = \{(1, 1), (-1, 1), (2, 4), (-2, 5)\}$ then two no different ordered pairs have the same first co-ordinates and therefore g is a function.

Domain and Range of a function:

Domain:- The domain is the set of all possible x -values (inputs)

Range: The range is the set of all possible y -values (outputs).

For example, consider the equation, $y = x^2$.
in the equation we assume that any real value of x is an allowable input since there is no restriction on x . Thus the equation defines a function $f(x) = x^2$ with domain $-\infty < x < \infty$.

As x varies over the domain of the function $f(x) = x^2$, the values of $y = x^2$ vary over the interval $0 \leq y < \infty$. So, this is the range of f .

Definition of different type of functions:

1. Single-valued function:

A function $y = f(x)$ is called a single-valued in a domain R if only one value of y corresponds to each value of x in R .

Example, $y = 2x + 5$, $y = x^2$, $y = e^{x+2}$, $y = \ln x + 5$ etc.

2. Many valued function:-

A function $y = f(x)$ is called a multivalued in a domain R if more than one value of y corresponds to each value of x in R .

For example, $y = \tan x$, $y = \sin^{-1} x$, $y = \cos^{-1} x$.

Suppose, $y = \sin^{-1} x$; $x = \frac{1}{2}$; $y = 30^\circ, 150^\circ, 390^\circ, \dots$ etc

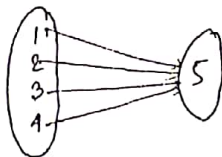
3. Explicit functions: A function which is directly expressed in terms of the independent variable is called an explicit function. i.e., $y = \cos x$, $y = x - 5x$ or $x = y - 2y + 1$.
4. Implicit function: A function which is not expressed directly in terms of the independent variable is called an implicit function.

$$x^3 + y^3 + 3xy = 0$$

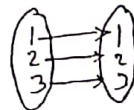
or, $F(x, y) = x^3 + 3x^2y + xy^2 + y^3 = 0$ [ind. & dep. mix]

5. Constant function: All elements of domain must be related with only one element of co-domain.

$$f(x) = 5 \text{ (constant)}$$



6. Identity function: If $f(x) = x$ then the function is called identity function.



7. Periodic function: If $f(x) = f(x+T)$ for all values of x then f is called the periodic function and T is called period. Trigonometrical functions such as $\sin x$ and $\cos x$ are periodic with 2π as period while

• Explicit: A f^n which is directly expressed in term of the independent variable is called an explicit f^n .
 e.g. $y = x \sin x$, $y = a \cos \theta$, $e^{ax} \cos bx$ etc.

• Implicit:
 $2x^2y^2 = a$
 $x^2 + 2x^2y + 4y^2 = 0$
 $3x^2 + 2x^2y + 4xy + 5x^2y^2 + 5 = 0$

$\tan x$ is periodic with period π .

$$\sin(2\pi + x) = \sin x \quad \& \quad \cos(2\pi + x) = \cos x$$

8. odd or even function A function $f(x)$ is said to be odd if $f(-x) = -f(x)$; and even if

$$f(-x) = f(x)$$

Example: 1. $f(x) = x^2$; $f(-x) = (-x)^2 = x^2 = f(x) = \text{even } f^?$

2. $f(x) = x^3$; $f(-x) = (-x)^3 = -x^3 = -f(x) = \text{odd } f^?$

Tip:- Trigonometric f^n : $y = \cos x$ & $y = \sec x$ are even functions, since, $y = f(x) = \cos x$
 $\Rightarrow y = f(-x) = \cos(-x) = \cos x = f(x) = \text{even}$

and $y = \sin x$
 $y = \csc x$
 $y = \tan x$
 $y = \cot x$ } odd f^n ; since $y = \sin(-x) = -\sin x$

Test whether the following functions are odd or even:

(i) $f(x) = 1 - \frac{1}{x^2}$

(ii) $g(x) = x + \frac{1}{x}$

(iii) $f(x) = \ln(x + \sqrt{1+x^2})$

Soln:-

(i) Given, $f(x) = 1 - \frac{1}{x^2}$ — (1)

putting $x = -x$ in eqn(1) we get,

$$f(-x) = 1 - \frac{1}{(-x)^2} = 1 - \frac{1}{x^2} = f(x) = \text{even fn}$$

(ii) Given $g(x) = x + \frac{1}{x}$ — (1)

putting $x = -x$ in (1)

$$g(-x) = (-x) + \frac{1}{(-x)} = -x - \frac{1}{x} = -(x + \frac{1}{x}) = -g(x)$$

Since $g(-x) = -g(x)$ therefore the given fn is odd fn.

(iii) $f(x) = \ln(x + \sqrt{1+x^2})$ — (1)

putting $x = -x$ in (1) we get

$$f(-x) = \ln(-x + \sqrt{1+(-x)^2})$$

$$= \ln(-x + \sqrt{1+x^2})$$

$$= \ln \left\{ \frac{(-x + \sqrt{1+x^2})(x + \sqrt{1+x^2})}{x + \sqrt{1+x^2}} \right\}$$

$$= \ln \left\{ \frac{(\sqrt{1+x^2})^2 - x^2}{x + \sqrt{1+x^2}} \right\}$$

$$= \ln \left(\frac{1}{x + \sqrt{1+x^2}} \right)$$

$$= -\ln(x + \sqrt{1+x^2}) = -f(x) = \text{odd fn}$$

One-one function:-

A function where denoted by $f: A \rightarrow B$ where

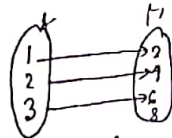
(i) Every domain has individual image in co-domain.

(ii) But co-domain may be empty.

is called one-one function.

Example:- The function $f(x) = 2x$

$$\text{then, } f: \{1, 2, 3\} \rightarrow \{2, 4, 6, 8\}$$



for $f(x) = 2x$ (or) $f(x) = 2x + 3$

Note: Co-domain:- A set containing the outputs called its co-domain.

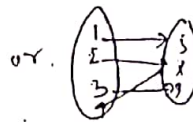
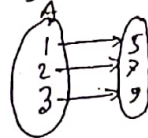
Onto function:- When a function i.e. (i) Every

co-domain must be related with domain.

(ii) But more than one domain cannot be related with same co-domain.

Suppose, $f = f(x) = 2x + 3$.

$$\text{then, } f: A = \{1, 2, 3\} \rightarrow B = \{5, 7, 9\}$$



for $f(x) = 2x + 3$
(or) $f(x) = 2x$
or $f(x) = 2x + 3$

Inverse function: If domain and co-domain are replaced to each other then it is called inverse function.

~~Suppose~~, $A = \{(-3, 5), (-2, 5), (-1, 5), (0, 5), (1, 5), (2, 5)\}$
 X : $A' = \{(5, -3), (5, -2), (5, -1), (5, 0), (5, 1), (5, 2)\}$

How do we find domain and Ranges.

Domain:

(i) If $f(x) = \frac{a}{0}$, $a \neq 0$, or $\frac{0}{0}$ or $a+ib$ ($b \neq 0$)

i.e., the denominator (bottom) of a fraction cannot be zero.

(ii) The number under a square root sign must be positive in this work. $f(x) = \sqrt{\phi(x)}$
 $\phi(x) \geq 0$

(iii) If $f(x) = \frac{\sqrt{\phi(x)}}{\sqrt{\psi(x)}}$ in this case

domain = $\phi(x) \geq 0$ and $\psi(x) \geq 0$.

Range:- the range is resulting y-values.

Find Domain and Range of the following functions:

(i) $f(x) = y = x + 3$ — (i)

Equation (i) satisfies all the real values of x .

So, Domain $f = D_f = \{x; x \in \mathbb{R}\} = \mathbb{R}$

From (i) $x = y - 3$ — (ii)

Again eqⁿ (ii) satisfies all the real values of x .

So, Range, $R_f = \{y; y \in \mathbb{R}\} = \mathbb{R}$.

$f(x) = 3x^2 + x + 2$ — (i)

Solⁿ:- Let $y = f(x) = 3x^2 + x + 2$ — (ii)

Eqⁿ (i) satisfies all the real values of x , so

Domain, $D_f = \{x; x \in \mathbb{R}\} = \mathbb{R}$.

From (i)

$$y = 3x^2 + x + 2$$

$$\Rightarrow 3x^2 + x + (2 - y) = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 3 \cdot (2 - y)}}{2 \cdot 3}$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{1 - 24 + 12y}}{6}$$
$$= \frac{-1 \pm \sqrt{12y - 23}}{6} \quad \text{--- (iii)}$$

Equation (iii) satisfies,

$$12y - 23 \geq 0$$

$$\text{i.e., } 12y \geq 23$$
$$\Rightarrow y \geq \frac{23}{12}$$

So, Range, $R_f = \{y: y \geq \frac{23}{12}\} = [\frac{23}{12}, \infty)$

$y = f(x) = \sqrt{x+3} \quad \text{--- (1)}$

$C_1^n(1)$ satisfied if and only if $x+3 \geq 0$

i.e. $x \geq -3$

So, Domain, $D_f = \{x: x \geq -3\} = [-3, \infty)$

Range, $R_f = \{y: y \geq 0\} = [0, \infty)$

$y = f(x) = \sqrt{16-x^2} \quad \text{--- (1)}$

$C_1^n(1)$ satisfies when $16-x^2 \geq 0$

$\Rightarrow x^2 \leq 16$

$\Rightarrow x \leq \pm 4$

$\Rightarrow -4 \leq x \leq 4$

Domain, $D_f = \{x: -4 \leq x \leq 4\} = [-4, 4]$

Range, $R_f = [0, 4]$

$y = \sqrt{4-x^2} \quad \text{--- (1)}$

$C_1^n(1)$ satisfies when $4-x^2 \geq 0$

$\Rightarrow x^2 \leq 4$

$\Rightarrow x \leq \pm 2$

$\Rightarrow -2 \leq x \leq 2$

Domain, $D_f = \{x: -2 \leq x \leq 2\} = [-2, 2]$

Range, $R_f = [0, 2]$

$y = \sqrt{x^2-16} \quad \text{--- (1)}$

Here, $x^2-16 \geq 0$ or, $x^2 \geq 16$. So, Domain consists of the intervals $x \leq -4$ and $x \geq 4$.

#

$$\# \checkmark f(x) = \sqrt{\frac{1-x}{x}} \quad \text{--- (1)}$$

eqn (1) valid only when $1-x \geq 0$ and $x > 0$

$$\therefore \text{Domain, } D_f = \{x : x > 0 \text{ and } x \leq 1\} = (0, 1]$$

$$\text{from (1) } y = \sqrt{\frac{1}{x} - 1}$$

$$x \in (0, 1) \text{ and } y \in [0, \infty)$$

$$\# f(x) = \frac{x-3}{2x+1} \quad \text{--- (1)}$$

$$\text{Domain, } D_f = \mathbb{R} - \{-\frac{1}{2}\}$$

Again from (1)

$$y = \frac{x-3}{2x+1}$$

$$\Rightarrow 2xy + y = x - 3$$

$$\Rightarrow 2xy - x = -y - 3$$

$$\Rightarrow x = \frac{-y-3}{2y-1}$$

$$\text{Range, } R_f = \mathbb{R} - \{\frac{1}{2}\}$$

$$\# y = f(x) = x^2 + 1$$

$$D_f = \mathbb{R}$$

$$R_f = [1, \infty)$$

$$x > 0 \Rightarrow f(x) = x^2 + 1$$

$$x < 0 \Rightarrow f(x) = x^2 + 1$$

$$x = 0 \Rightarrow f(x) = 1$$

$$\text{H.W.:- } y = f(x) = \frac{x-1}{2x-3} \quad [\text{from H.S.C. text book}]$$

$$f(x) = \frac{x}{|x|}$$

Soln: Given, $f(x) = \frac{x}{|x|}$

If $x > 0$, $f(x) = \frac{0}{101} = \frac{0}{0}$ which is undefined

Domain of this function, $D_f = \mathbb{R} - \{0\}$

Range: $f(x) = \begin{cases} \frac{x}{x} & \text{if } x > 0 \\ \frac{x}{-x} & \text{if } x < 0 \end{cases}$

$$= \begin{cases} 1 & \text{when } x > 0 \\ -1 & \text{when } x < 0 \end{cases}$$

Range of this f , $R_f = \{1, -1\}$

$f(x) = |x| - x$

$$f(x) = \begin{cases} x - x & \text{when } x \geq 0 \\ -x - x & \text{when } x < 0 \end{cases}$$

$$= \begin{cases} 0 & \text{when } x \geq 0 \\ -2x & \text{when } x < 0 \end{cases}$$

Domain: $D_f = [0, \infty) \cup (-\infty, 0)$

$$= (-\infty, \infty) = \mathbb{R}$$

Range: For $[0, \infty)$, $f(x) = 0$, $R_f = \{0\}$

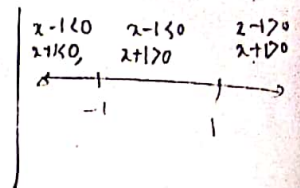
For $(-\infty, 0)$, $f(x) = -2x$, $R_f = (0, \infty)$

Range of this f , $R_f = \{0\} \cup (0, \infty) = [0, \infty)$

Find domain and Range of $f(x) = |x+1| + |x-1|$

Soln:- Given, $f(x) = |x+1| + |x-1|$

$$\therefore f(x) = \begin{cases} (x+1) + (x-1) & \text{when } x \geq 1 \\ (x+1) - (x-1) & \text{when } -1 \leq x < 1 \\ -(x+1) - (x-1) & \text{when } x < -1 \end{cases}$$



$$= \begin{cases} 2x & \text{when } x \geq 1 \\ 2 & \text{when } -1 \leq x < 1 \\ -2x & \text{when } x < -1 \end{cases}$$

Domain of this f^n , $D_f = [1, \infty) \cup [-1, 1) \cup (-\infty, -1)$
 $= (-\infty, \infty) = \mathbb{R}$

Range: For, $[1, \infty)$; $f(x) = 2x$, $R_f = [2, \infty)$

For, $[-1, 1)$; $f(x) = 2$, $R_f = \{2\}$

For, $[-1, -\infty)$; $f(x) = -2x$, $R_f = (2, \infty)$

$\therefore$ Range of this f^n ; $R_f = [2, \infty) \cup \{2\} \cup (2, \infty)$
 $= [2, \infty)$

#1 $f(x) = |x| + |x-1|$

Soln:-



$$f(x) = |x| + |x-1|$$

$$= \begin{cases} -x - (x-1) & \text{when } x < 0 \\ x - (x-1) & \text{when } 0 \leq x < 1 \\ x + x - 1 & \text{when } x \geq 1 \end{cases}$$

$$= \begin{cases} -2x + 1 & \text{when } x < 0 \\ 1 & \text{when } 0 \leq x < 1 \\ 2x - 1 & \text{when } x \geq 1 \end{cases}$$

$$\text{Domain, } D_f = (-\infty, 0) \cup [0, 1) \cup [1, \infty)$$

$$= (-\infty, \infty) = \mathbb{R}$$

$$R_f = [1, \infty) \cup \{1\} \cup [1, \infty)$$

$$= [1, \infty)$$

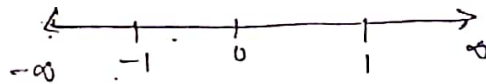
$f(x) = |x-1| - |x-2| \rightarrow \text{H.W.}$

Hint: $D_f = \mathbb{R}$

$$R_f = [1, 1]$$

$$f(x) = |x+1| + |x| + |x-1|$$

Soln:



$$= \begin{cases} -(x+1) - x - (x-1) & ; x < -1 \\ x+1 - x - (x-1) & ; -1 \leq x < 0 \\ (x+1) + x - (x-1) & ; 0 \leq x < 1 \\ x+1 + x + x-1 & ; x \geq 1 \end{cases}$$

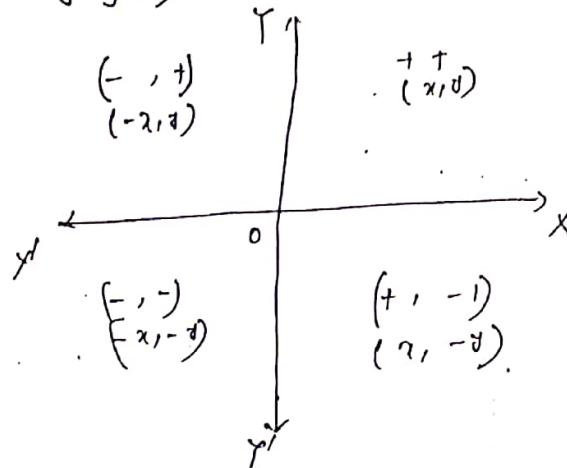
$$= \begin{cases} -3x & ; x < -1 \\ -x+2 & ; -1 \leq x < 0 \\ x+2 & ; 0 \leq x < 1 \\ 3x & ; x \geq 1 \end{cases}$$

$$\text{Domain, } D_f = (-\infty, -1) \cup [-1, 0) \cup [0, 1) \cup [1, \infty) \\ = (-\infty, \infty) = \mathbb{R}$$

$$R_f = [3, \infty) \cup (2, 3] \cup [2, 3) \cup [3, \infty) \\ = [2, \infty)$$

Graph, Domain and Range of a function

Graph of a function:- The graph of a function $y=f(x)$ is the locus of all points (x,y) satisfied by the equation $y=f(x)$.

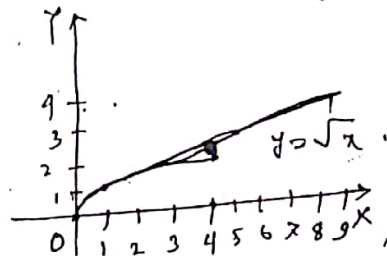


Graph of Some common fⁿ:

(i) $y = f(x) = \sqrt{x}$

Solⁿ:

x	0	1	4	9
y	0	1	2	3



(ii) $y = \frac{1}{x}$
 $\therefore y = f(x) = \frac{1}{x}$ undefined
 $\therefore f(0) = \frac{1}{0}$
 $y = \frac{1}{x}; x < 0$

x	-2	-1	-1/2	-1/3
y	-1/2	-1	-2	-3

$y = \frac{1}{x}; x > 0$				
x	1/3	1/2	1	2
y	3	2	1	1/2

Find the domain and range of

$$f(x) = \begin{cases} 5 & ; 0 < x < 1 \\ 10 & ; 1 < x \leq 2 \\ 15 & ; 2 < x \leq 3 \end{cases}$$

$$D_f = (0, 1) \cup (1, 2] \cup (2, 3]$$

$$= (0, 3] - \{1\}$$

Range: $R_f = \{5, 10, 15\}$

Assignment

Draw the graphs of the functions, and find their domains and ranges

(i) $f(x) = -x^2 + 1$; $D_f = \mathbb{R}$, $R_f: y \leq 1$

(ii) $f(x) = \begin{cases} x-1 & ; 0 < x < 1 \\ 2x & ; 1 \leq x \end{cases} \rightarrow D_f: x > 0 \text{ \& } R_f: -1 < y < 0 \text{ or } y \geq 2$

(iii) $y = f(x) = \frac{x^2-1}{x-2}$; $D_f: x \neq 2$, $R_f: y \neq 4$

(iv) $y = f(x) = \frac{1}{x}$; $D_f: x \neq 0$, $R_f: y \neq 0$

(v) $y = f(x) = x - |x|$; $D_f = \mathbb{R}$, $R_f: y \leq 0$

(vi) $f(x) = \sqrt{\frac{x+1}{x-1}}$; $D_f = (-\infty, -1) \cup (1, \infty)$
 $\& R_f = \mathbb{R} - \{-1, 1\}$

Draw the graph of the function defined by

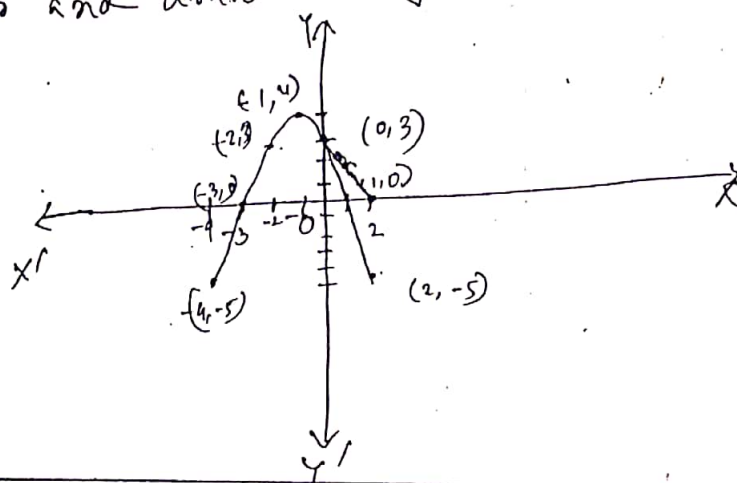
$$y = 3 - 2x - x^2$$

Solⁿ:- let $y = f(x) = 3 - 2x - x^2$

Now we draw the following table.

x	-5	-4	-3	-2	-1	0	1	2	3	4
y	-12	-5	0	3	4	3	0	-5	-12	-21

From the table we plot the point in the graph papers and draw the graph through this point.



✓ # let $y = f(x) = \begin{cases} x^2 + 1 & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ 1/x & \text{when } 1 < x \end{cases}$

Solⁿ

Now we draw the following tables.

(i)

$$y = x^2 + 1 \quad ; \quad x < 0$$

x	-1	$-x_2$	$-1/4$
y	2	$5/4$	$17/16$

(ii)

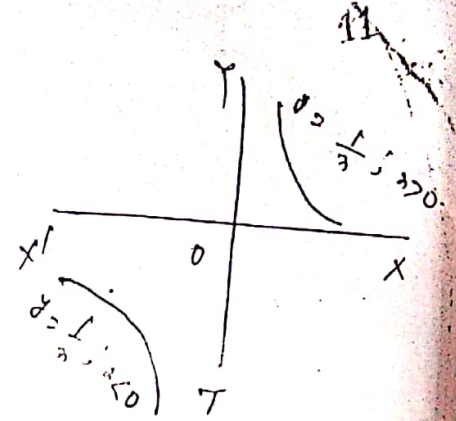
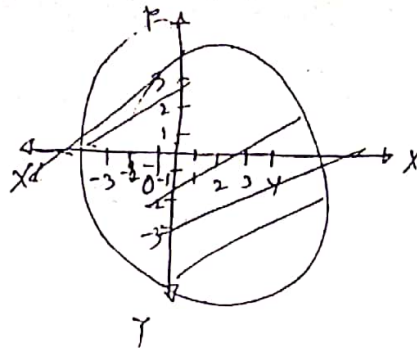
$$y = x \quad ; \quad 0 \leq x \leq 1$$

x	0	x_2	1
y	0	x_2	1

(iii)

x	$1/3$	2	3
y	$3/4$	x_2	$1/3$

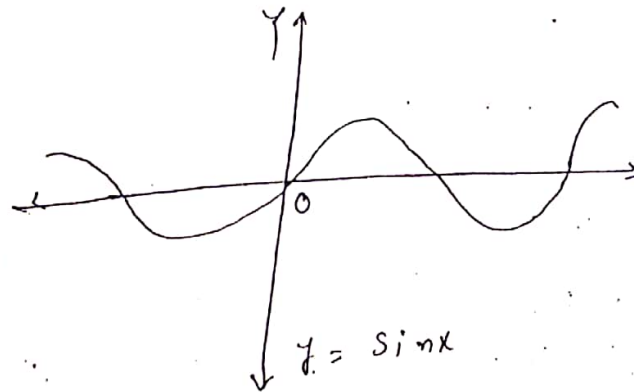
From the table we plot the point in the graph papers and draw the graph through this point.



* $y = \sin x$
Now we draw the following table

x	$-\pi$	$-\pi/2$	0	$\pi/2$	π	$3\pi/2$	2π
y	0	-1	0	1	0	-1	0

From the table we plot the point in the graph paper and draw the graph through this point.

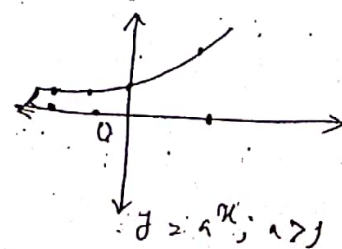


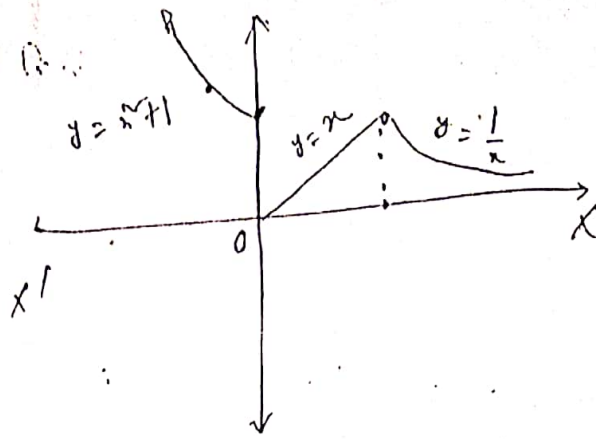
Here, $D_f = \mathbb{R}$ & $R_f = [-1, 1]$

Similarly, $\cos x$, $\tan x$

$y = f(x) = a^x$; $a > 1$.

x	-3	-2	-1	0	1	2
y	y_8	y_4	y_2	1	2	4





Here, $D_f = (-\infty, 0) \cup [0, 1] \cup (1, \infty)$

$\sup = (-\infty, 0)$

$R_f = [1, \infty) \cup [0, 1] \cup (0, 1)$

$\inf = [0, \infty)$

7/ $y = |x| + |x-1|$ [previous problem]

Soln:

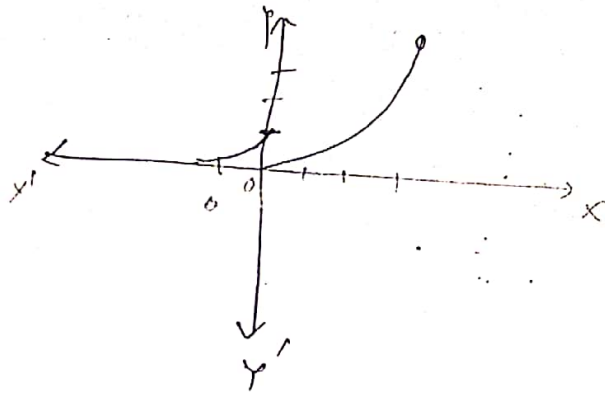


Given, $y = f(x) = |x| + |x-1|$

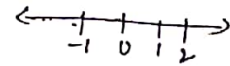
$$= \begin{cases} -2x+1 & ; x < 0 \\ 1 & ; 0 \leq x < 1 \\ 2x-1 & ; x \geq 1 \end{cases}$$

Now, we draw the following table:

① $y = -2x+1, x < 0$				② $y = 1, 0 \leq x < 1$			③ $y = 2x-1, 1 \leq x$			
x	-2	-1	-1/2	x	0	1/2	x	1	2	3
y	5	3	2	y	1	1	y	1	3	5



Here, $D_f = (-1, 0) \cup [0, 2)$
 $= (-1, 2)$



and $R_f = (e^{-1/2}, 1) \cup [0, 4)$
 $= [0, 4)$

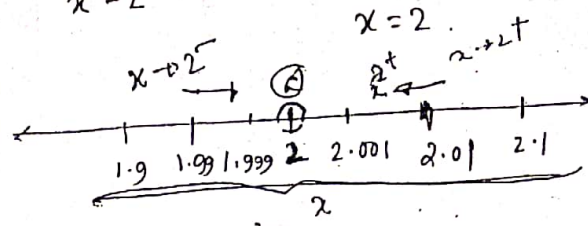
Assignment:- Draw the graph and find domain & Range.

(i)
$$f(x) = \begin{cases} 3+2x & ; -3/2 < x < 0 \\ 3-2x & ; 0 \leq x < 3/2 \\ 3+2x & ; 3/2 \leq x \end{cases}$$

(ii)
$$f(x) = \begin{cases} 0 & ; |x| > 1 \\ 1+x & ; -1 \leq x \leq 0 \\ 1-x & ; 0 < x \leq 1 \end{cases}$$

1F(3)

* $f(x) = \frac{x^2 - 4}{x - 2}$ this function is undefined when $x = 2$.



$x = 1.99, f(x) = 3.99$
 $x = 1.999, f(x) = 3.999$
 $x = 2.001, f(x) = 3.999$
 $x = 2.01, f(x) = 4.01$
 $x = 2.1, f(x) = 4.41$

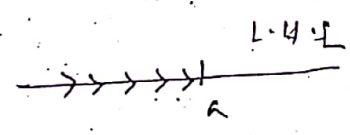
$$f(1.999) = \frac{(1.999)^2 - 4}{1.999 - 2} = 3.999 \quad (L.H.L)$$

$$f(2.001) = \frac{(2.001)^2 - 4}{2.001 - 2} = 4.001 \quad (R.H.L)$$

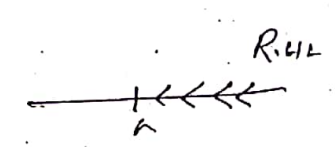
$$\begin{aligned}
 f(x) &= \frac{x^2 - 4}{x - 2} \\
 &= \frac{(x+2)(x-2)}{(x-2)} \\
 &= x+2 \quad [x \neq 2] \\
 &= 4 \quad (L)
 \end{aligned}$$

$x \rightarrow 2$ (x এর দিকে)
 যখন x এর দিকে
 গিয়েছে তখন
 $|x-2| \rightarrow 2+5$
 x এর দিকে গিয়েছে তখন
 x এর দিকে গিয়েছে তখন
 x এর দিকে গিয়েছে তখন

L.H.L (Left hand Limit)
 $x \rightarrow 2^-$



R.H.L (Right hand Limit)



Mir@cle Point
 আলোয়া ফটোগ্রাফির দিকে
 কোম্পানির সামনে, কলকাতা রোড, চকবাজার, চট্টগ্রাম।
 IIUC এর সকল লেকচার শীট ও পূর্ববর্তী
 সেমিটারের প্রশ্ন পাওয়া যায়।
 মোবাইল নং ০১৭৭৫৯৪৩২৬৯

Signature

Limit of a function:-

When the variable x in approaches the fixed number ' a ' from both sides ultimately becomes very near equal to ' a '. If $f(x)$ approaches a fixed number ' l ', then ' l ' is said to be the limit of $f(x)$.

This is expressed by writing $\lim_{x \rightarrow a} f(x) = l$

$\lim_{x \rightarrow a^+} f(x)$ is called Right hand limit

$\lim_{x \rightarrow a^-} f(x)$ is called Left hand limit.

P-1: Given, $f(x) = \frac{x^2 - 5x + 6}{x - 3}$; does $\lim_{x \rightarrow 3} f(x)$ exist?

Soln: Given, $f(x) = \frac{x^2 - 5x + 6}{x - 3}$

$$= \frac{(x-3)(x-2)}{x-3}$$

L.H.L.,

$$\lim_{x \rightarrow 3^-} f(x)$$

$$= \lim_{x \rightarrow 3^-} (x-2)$$

$$= 3-2$$

$$= 1$$

$$= x-2$$

$$R.H.L = \lim_{x \rightarrow 3^+} f(x)$$

$$= \lim_{x \rightarrow 3^+} (x-2)$$

$$= 3-2$$

$$= 1$$

Since $L.H.L = R.H.L$; therefore $\lim_{x \rightarrow 3} f(x)$ exist.

Q. 11. w Given $f(x) = \frac{x^2 - 9}{x - 3}$ does $\lim_{x \rightarrow 3} f(x)$ exist.

Q. 12. If $f(x) = \begin{cases} 1+2x & \text{when } -\frac{1}{2} \leq x < 0 \\ 1-2x & \text{when } 0 \leq x < \frac{1}{2} \\ -1+2x & \text{when } x > \frac{1}{2} \end{cases}$

does $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow \frac{1}{2}} f(x)$ exist?

Soln: Given, $f(x) = \begin{cases} 1+2x & \text{when } -\frac{1}{2} \leq x < 0 \\ 1-2x & \text{when } 0 \leq x < \frac{1}{2} \\ -1+2x & \text{when } x > \frac{1}{2} \end{cases}$

$$L.H.L = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (1+2x) = 1+0 = 1$$

$$R.H.L = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (1-2x) = 1-0 = 1$$

Since $L.H.L = R.H.L$, therefore $\lim_{x \rightarrow 0} f(x)$ exist.

Again, $L.H.L = \lim_{x \rightarrow \frac{1}{2}^-} f(x) = \lim_{x \rightarrow \frac{1}{2}^-} (1-2x) = 1-2 \cdot \frac{1}{2} = 0$

$$R.H.L = \lim_{x \rightarrow \frac{1}{2}^+} f(x) = \lim_{x \rightarrow \frac{1}{2}^+} (-1+2x) = -1+2 \cdot \frac{1}{2} = 0$$

Since $L.H.L = R.H.L$, therefore $\lim_{x \rightarrow \frac{1}{2}} f(x)$ exist.

④ ^{H.W} let, the function $f(x)$ is defined as follows:

$$f(x) = \begin{cases} x^n & \text{when } x < 1 \\ 0 & \text{when } x = 1 \\ x^{n+1} & \text{when } x > 1 \end{cases}$$

Does $\lim_{x \rightarrow 1} f(x)$ exist?

Soln: Given, $f(x) = \begin{cases} x^n & \text{when } x < 1 \\ 0 & \text{when } x = 1 \\ x^{n+1} & \text{when } x > 1 \end{cases}$

$$\text{L.H.L} \Rightarrow \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^n = 1^n = 1$$

$$\text{R.H.L} \Rightarrow \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} x^{n+1} = 1^{n+1} = 2$$

Since $\text{L.H.L} \neq \text{R.H.L}$,

therefore $\lim_{x \rightarrow 1} f(x)$ doesn't exist.

⑤ let, $f(x) = \frac{3x + |x|}{x - 5|x|}$ does $\lim_{x \rightarrow 0} f(x)$ exist?

Soln: Given $f(x) = \frac{3x + |x|}{x - 5|x|}$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{3x - x}{x + 5x} = \frac{2x}{12x} = \frac{1}{6} \\ &\Rightarrow \lim_{x \rightarrow 0^-} \frac{3x + |x|}{x - 5|x|} \end{aligned}$$

$$\text{R.H.L, } \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{3x+1}{x-5/x}$$

$$= \frac{3x+x}{x-5x}$$

$$= \frac{4x}{-4x} = -1$$

Since, L.H.L $\neq$ R.H.L, therefore, $\lim_{x \rightarrow 0} f(x)$ doesn't exist.

Q. Given, $f(x) = \begin{cases} x & \text{when } x > 0 \\ 0 & \text{when } x = 0 \\ -x & \text{when } x < 0 \end{cases}$

It is

does $\lim_{x \rightarrow 0} f(x)$ exist?

Soln:- Given, $f(x) = \begin{cases} x & \text{when } x > 0 \\ 0 & \text{when } x = 0 \\ -x & \text{when } x < 0 \end{cases}$

$$\text{L.H.L} = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} -x = 0$$

$$\text{R.H.S, } = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0.$$

Since, L.H.L = R.H.L.

therefore, $\lim_{x \rightarrow 0} f(x)$ exist.

Continuity of a function

Continuity:- A function $f(x)$ is said to be continuous for $x=a$, provided $\lim_{x \rightarrow a} f(x)$ exists, is finite, and is equal to $f(a)$

In other words, for $f(x)$ to be continuous at $x=$

$$\lim_{x \rightarrow a+0} f(x) = \lim_{x \rightarrow a-0} f(x) = f(a)$$

Ex:- $f(x) = x^n$ is continuous for any value of x for,

Discontinuity of a function $\lim_{x \rightarrow a} x^n = a^n$

A function which is not continuous at point is said to have a discontinuity at that point.

not

Ex:- $f(x) = \frac{1}{x^n}$ is discontinuous at $x=0$, for

$\lim_{x \rightarrow 0} \left(\frac{1}{x^n}\right)$ is not finite.

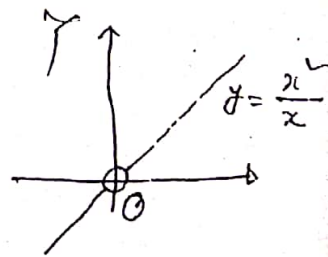
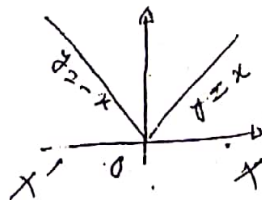
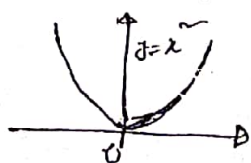
or,

$f(x) = \cos \frac{1}{x}$ is discontinuous at $x=0$,

since $\lim_{x \rightarrow 0} \cos \frac{1}{x}$ does not exist.

Ex:-

$$y = x^2$$



Test the continuity of the following function.

at $x=2$,

$$f(x) = \begin{cases} 5 & \text{when } 0 < x < 1 \\ 10 & \text{when } 1 < x \leq 2 \\ 15 & \text{when } 2 < x \leq 3. \end{cases}$$

Soln:-

When $x=2$

$$\Rightarrow f(x) = 10$$

$$\Rightarrow f(2) = 10$$

Soln

$$\text{L.H.L.} = \lim_{x \rightarrow 2^-} f(x)$$

$$= \lim_{x \rightarrow 2^-} 10$$

$$= 10$$

$$\text{R.H.L.} = \lim_{x \rightarrow 2^+} f(x)$$

$$= \lim_{x \rightarrow 2^+} 15$$

$$= 15$$

Since $\text{L.H.L.} \neq \text{R.H.L.}$, therefore $\lim_{x \rightarrow 2} f(x)$ doesn't exist.

The function is not continuous at point $x=2$.

Test the continuity of the following function at $x=0$ and $x=\frac{3}{2}$,

$$f(x) = \begin{cases} 3+2x & \text{when } -\frac{3}{2} \leq x \leq 0 \\ 3-2x & \text{when } 0 \leq x \leq \frac{3}{2} \\ -3-2x & \text{when } x > \frac{3}{2} \end{cases}$$

4

Soln:-

$$\text{Given, } f(x) = \begin{cases} 3+2x & \text{when } -\frac{3}{2} \leq x < 0 \\ 3-2x & \text{'' } 0 \leq x < \frac{3}{2} \\ -3-2x & \text{'' } x \geq \frac{3}{2} \end{cases}$$

When, $x \geq 0$, $f(x) = 3-2x$

$$f(0) = 3 - 0 = 3$$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 0^-} f(x) \\ &= \lim_{x \rightarrow 0^-} (3+2x) \\ &= 3+0 \\ &= 3 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 0^+} f(x) \\ &= \lim_{x \rightarrow 0^+} (3-2x) \\ &= 3-0 \\ &= 3 \end{aligned}$$

Since, L.H.L = R.H.L, therefore, $\lim_{x \rightarrow 0} f(x)$ ^{does exist.}

Again

when, $x = \frac{3}{2}$, $f(x) = -3-2x$

$$\Rightarrow f\left(\frac{3}{2}\right) = -3 - 2 \cdot \frac{3}{2} = -6$$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow \frac{3}{2}^-} f(x) \\ &= \lim_{x \rightarrow \frac{3}{2}^-} (3-2x) \\ &= 3 - 2 \cdot \frac{3}{2} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow \frac{3}{2}^+} f(x) \\ &= \lim_{x \rightarrow \frac{3}{2}^+} (-3-2x) \\ &= -3 - 2 \cdot \frac{3}{2} \\ &= -6 \end{aligned}$$

Since, L.H.L $\neq$ R.H.L, therefore $\lim_{x \rightarrow \frac{3}{2}} f(x)$ doesn't exist.

So the function is not continuous at the point $x = \frac{3}{2}$ but continuous at $x = 0$.

Assignment

③ Test the continuity at $x=0, x=1$

$$f(x) = \begin{cases} x^2 + 1 & \text{when } x < 0 \\ x & \text{when } 0 \leq x \leq 1 \\ 1/x & \text{when } x > 1 \end{cases}$$

④ Test the continuity at $x=0, 1, 2$

Assignment

$$f(x) = \begin{cases} -x^2 & \text{when } x \leq 0 \\ 5x - 4 & \text{when } 0 < x \leq 1 \\ 4x^2 - 3x & \text{when } 1 < x < 2 \\ 3x + 4 & \text{when } x \geq 2 \end{cases}$$

⑤ Test the continuity at $x=0$

$$f(x) = \begin{cases} \frac{\ln x}{x} & \text{when } x \neq 0 \\ 1 & \text{when } x = 0. \end{cases}$$

soln:- when, $x=0$, $f(x) = 1 \Rightarrow f(0) = 1$

$$\text{L.H.L.} \quad \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{x \rightarrow 0^-} \frac{|x|}{x}$$

$$= \frac{-x}{x}$$

$$= -1$$

$$\text{R.H.L.} \quad \lim_{x \rightarrow 0^+} f(x)$$

$$= \lim_{x \rightarrow 0^+} \frac{|x|}{x}$$

$$= \frac{x}{x} = 1$$

Since L.H.L. $\neq$ R.H.L., therefore $\lim_{x \rightarrow 0}$ doesn't exist.

So the function is not continuous at $x = 0$.

(6) Test the continuity at $x = 0$

$$f(x) = \begin{cases} -\cos x & \text{when } x \neq 0 \\ \cos x & \text{when } x = 0 \end{cases}$$

(7) Test the continuity at $x = 0$.

$$f(x) = \begin{cases} \frac{\log_e(1+x)}{\tan^{-1}x} & \text{when } x \neq 0 \\ 1 & \text{when } x = 0 \end{cases}$$

Soln: Given $f(x) =$
 when $x = 0$, $f(x) = 1$
 $f(0) = 1$

$$L.H.L. = \lim_{x \rightarrow 0^-} f(x)$$

$$= \lim_{x \rightarrow 0^-} \frac{\log_e(1+x)}{\tan^{-1} x}$$

$$= \lim_{x \rightarrow 0^-} \frac{(1 - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots)}{(x - \frac{x^3}{3} + \frac{x^5}{5} - \dots)}$$

$$= \lim_{x \rightarrow 0^-} \frac{x(1 - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots)}{x(1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots)}$$

$$= \lim_{x \rightarrow 0^-} \frac{1 - \frac{x^2}{2} + \frac{x^3}{3} - \dots}{1 - \frac{x^2}{3} + \frac{x^4}{5} - \dots}$$

$$= \frac{1}{1} = 1$$

$$R.H.L. = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\log_e(1+x)}{\tan^{-1} x} = 1$$

$$\therefore \lim_{x \rightarrow 0} f(x) \text{ exist.}$$

Since $\lim_{x \rightarrow 0} f(x) = f(0)$; therefore the function is continuous at $x=0$.

* Test the continuity at $x=1, 2$

$$f(x) = \begin{cases} \log x & \text{when } 0 < x \leq 1 \\ 0 & \text{when } 1 < x < 2 \\ 1+x & \text{when } x > 2 \end{cases}$$

when,

$$x = 1$$

$$f(x) = \log x$$

$$\Rightarrow f(1) = \log(1) = 0$$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 1^-} f(x) \\ &= \lim_{x \rightarrow 1^-} \log x \\ &= \log(1) \\ &= 0 \end{aligned}$$

R.H.S,

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) \\ &= \lim_{x \rightarrow 1^+} 0 \\ &= 0 \end{aligned}$$

$$\therefore \lim_{x \rightarrow 1} f(x) \text{ exist.}$$

Since $\lim_{x \rightarrow 1} f(x) = f(1)$ therefore the function is continuous at $x=1$.

$$\text{When } x=2, f(x)=0, \Rightarrow f(2)=0.$$

$$\begin{aligned} \text{L.H.L} &= \lim_{x \rightarrow 2^-} f(x) \\ &= \lim_{x \rightarrow 2^-} 0 = 0 \end{aligned}$$

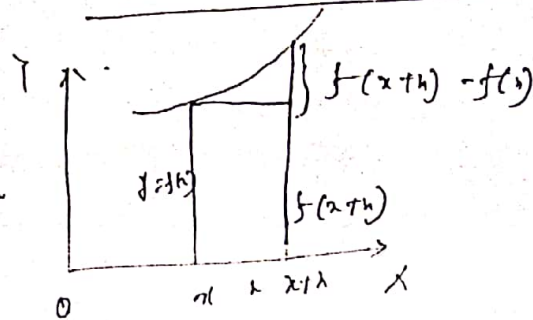
$$\begin{aligned} \text{R.H.L} &= \lim_{x \rightarrow 2^+} f(x) \\ &= \lim_{x \rightarrow 2^+} (1+x) \end{aligned}$$

Since, L.H.L $\neq$ R.H.L

So the $\lim_{x \rightarrow 2} f(x)$ does not exist.

Therefore the function is not continuous at $x=2$.

Differentiability of a f



$$\text{Slope} = \frac{dy}{dx} = \frac{f(x+h) - f(x)}{x+h - x} = \frac{f(x+h) - f(x)}{h}$$

if $h \rightarrow 0$

$$\frac{dy}{dx} \text{ or } y' \text{ or } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Condition:

Let $f(x)$ be a function which is defined in the interval $[a, b] \subset \mathbb{R}$ and let $a < c < b$ then the function $f(x)$ is said to be differentiable at $x=c$ if $f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$ exist?

and $Rf'(c) = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}$

and $Lf'(c) = \lim_{h \rightarrow 0^-} \frac{f(c-h) - f(c)}{-h}$ are

finite and equal $[Lf'(c) = Rf'(c)]$

* Test whether the $f^n f(x) = x^2 + 5$ is differentiable at the point $x=1$.

$$Rf'(1) = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$\left| \begin{array}{l} f(1) = 1 + 5 = 6 \\ f(1+h) = (1+h)^2 + 5 \end{array} \right.$$

$$\therefore Rf'(1) = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{(1+h)^2 + 5 - 6}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{1 + 2h + h^2 + 5 - 6}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{2h + h^2}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{h(2+h)}{h}$$

$$= 2 + 0 = 2$$

$$Lf'(1) = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$Lf'(1) = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{1 + 2h + h^2 + 5 - 6}{h}$$

$$= \lim_{h \rightarrow 0^-} (2+h) = 2 + 0 = 2$$

Since $Lf'(1) \stackrel{h \rightarrow 0^-}{=} Rf'(1)$ and both are finite so the f^n is differentiable at $x=1$

A fn $f(x)$ is defined as
when $0 < x \leq 1$

$$f(x) = \begin{cases} 5x+4 & \text{when } 0 < x \leq 1 \\ 4x^2-3x & \text{when } 1 < x < 2 \end{cases}$$

① Test the continuity at $x=1$ - H.W

② Test whether the above f^n is differentiable at $x=1$

Soln: $R f'(1) = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$

$$\begin{aligned} & \lim_{h \rightarrow 0^+} \frac{4(1+h)^2 - 3(1+h) - 1}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{4(1+2h+h^2) - 3-3h-1}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{4+8h+4h^2-3-3h-1}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{4h+5h^2}{h} \\ &= \lim_{h \rightarrow 0^+} (4+5h) \\ &= 5 \end{aligned}$$

iff $x > 1$
 $\Rightarrow x-1 > 0$
 $\Rightarrow h > 0$ i.e. $h \rightarrow 0^+$
 $f(x) = 4x^2 - 3x$
 $f(1) = 4 - 3 = 1$
 $f(1+h) = 4(1+h)^2 - 3(1+h)$
 $= 4(1+2h+h^2) - 3-3h$
 $= 4+8h+4h^2-3-3h$
 $= 1+5h+4h^2$

$$L f'(1) = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$\begin{aligned} &= \lim_{h \rightarrow 0^-} \frac{5+5h-5}{h} \\ &= \lim_{h \rightarrow 0^-} (5) \\ &= 5 \end{aligned}$$

iff
 $x < 1, f(x) = 5x+4$
 $x-1 < 0$
 $h < 0 \Rightarrow h \rightarrow 0^-$
 $f(1) = 5+4 = 9$
 $f(1+h) = 5(1+h) + 4$
 $= 5+5h+4$
 $= 9+5h$

Since $Rf'(1) = Lf'(1)$ and both are finite
 so the f is differentiable at $x=1$.

68.38

① when

$$f(x) = 3x$$

$$= 3 \cdot 1$$

$$1 = 3 \cdot 1$$

②

$$f(x) = 4x$$

$$1 = 4 \cdot 1$$

$$1 = 4 \cdot 1$$

$$1 = 4 \cdot 1$$

More problems: From recommended books.

$$x = 0.5$$

$$y = 1.5$$

$$z = 1$$

$$x = 1.5$$

$$y = 1.5$$

$$z = 1.5$$

$$x = 1.5$$

$$y = 1.5$$

$$z = 1.5$$

$$x = 1.5$$

$$y = 1.5$$

$$z = 1.5$$

$$x = 1.5$$

$$y = 1.5$$

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$$z = 1.5$$

$$x = 1.5$$

$$y = 1.5$$

$$z = 1.5$$

$$x = 1.5$$

$$y = 1.5$$

$$z = 1.5$$

$$x = 1.5$$

$$y = 1.5$$

$$z = 1.5$$

$$x = 1.5$$

$$\text{H.L.} \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^+} 2 + \left(x - \frac{\pi}{2}\right)^{\sqrt{\quad}}$$

$$= \lim_{x \rightarrow \frac{\pi}{2}^+} 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^{\sqrt{\quad}} = 2 + 0 = 2$$

L.H. When $x < \frac{\pi}{2}$ then $f(x) = 1 + \sin x$

$$\text{L.H.L.} = \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} 1 + \sin x = 1 + \sin \frac{\pi}{2} = 1$$

When $x = \frac{\pi}{2}$ then $f(x) = 2 + \left(x - \frac{\pi}{2}\right)^{\sqrt{\quad}}$

$$f\left(\frac{\pi}{2}\right) = 2 + \left(\frac{\pi}{2} - \frac{\pi}{2}\right)^{\sqrt{\quad}} = 2$$

$$\text{Since, } \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = f\left(\frac{\pi}{2}\right)$$

Therefore, $x = \frac{\pi}{2}$ $f(x)$ is continuous.

For differentiability: $f(x)$ is continuous at $x = \pi/2$

$$Rf'(x) = \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$$

$$\therefore Rf'\left(\frac{\pi}{2}\right) = \lim_{h \rightarrow 0^+} \frac{f\left(\frac{\pi}{2}+h\right) - f\left(\frac{\pi}{2}\right)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{2 + \left(\frac{\pi}{2}+h - \frac{\pi}{2}\right)^{\sqrt{\quad}} - 2}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{h^{\sqrt{\quad}}}{h} = \lim_{h \rightarrow 0^+} h^{\sqrt{\quad}-1} = 0$$

When $x < \frac{\pi}{2}$ then $f(x) = 1 + \sin x$

$$\therefore f\left(\frac{\pi}{2}+0\right) - f\left(\frac{\pi}{2}\right) = 1 + \sin\left(\frac{\pi}{2}\right) - 1 = 1$$

$$\begin{aligned}
 2 \lim_{h \rightarrow 0^+} \frac{\cos h - 1}{h} &= \lim_{h \rightarrow 0^+} \frac{(1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots)}{h} \quad [\cos h = 1 - \frac{h^2}{2!}] \\
 &= \lim_{h \rightarrow 0^+} \frac{-\frac{h^2}{2!} + \frac{h^4}{4!} - \dots}{h} \\
 &= \lim_{h \rightarrow 0^+} 0 = 0
 \end{aligned}$$

$\therefore Lf'(\frac{\pi}{2}) = Rf'(\frac{\pi}{2})$ therefore the $f(x)$ is differentiable

* (ϵ, δ) defn of limit $\rightarrow \lim_{x \rightarrow a} f(x) = l$.
 Given a number $\epsilon > 0$, however small, there exists a positive number δ such that for $x(x) \in \delta > 0$ satisfying $|x - a| < \epsilon$

$$\Rightarrow |f(x) - l| < \epsilon$$

1. prove that $\rightarrow \lim_{x \rightarrow 2} (2x+1) = 5$
 $f(x) = 5$

Let us consider an arbitrary positive number $\epsilon > 0$, however small.
 then $|f(x) - 5| = |2x+1 - 5| < \epsilon \quad |2x-4| < \epsilon$

$$\Rightarrow |f(x) - 5| < \epsilon \quad \Rightarrow |2(x-2)| < \epsilon$$

$$|x-2| < \frac{\epsilon}{2}$$

$$\Rightarrow |x-2| < \delta \quad \left| \delta = \frac{\epsilon}{2} > 0 \right.$$

this means that $\lim_{x \rightarrow 2} (2x+1) = 5$

A f^n is defined as $f(x) = \begin{cases} 5x-4 & \text{when } 0 < x < 1 \\ 4x^2-3x & \text{!! } 1 < x < 2 \end{cases}$

- (i) Does limit exist $x \rightarrow 1$
 (ii) Test the continuity at $x=1$
 (iii) Test whether the above f^n is differentiable at $x=1$.

(ii) When $x \neq 1$ then $f(x) = 5x-4$

$$\text{L.H.L} = \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (5x-4) = 5 \cdot 1 - 4 = 1$$

$$\text{R.H.L} = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (4x^2-3x) = 4 \cdot 1^2 - 3 \cdot 1 = 1$$

When $x = 1$ ✓

$$\begin{aligned} f(1) &= 5x-4 \\ &= 5 \cdot 1 - 4 \\ &= 1 \end{aligned}$$

Since, L.H.L = R.H.L, therefore $\lim_{x \rightarrow 1} f(x)$ does.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\text{if, } \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x-h) - f(x)}{h} = \text{L.H.L} = \text{R.H.L}$$

the $f^n f(x)$ is differentiable at

$$\text{L.H.L} = \text{R.H.L}$$

so/:-

$$Rf'(1) = \lim_{h \rightarrow 0^+} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4(1+h)^2 - 3(1) - (4 \cdot 1^2 - 3 \cdot 1)}{h}$$

$$= \lim_{h \rightarrow 0^+} \frac{4(1+h)^2 - 3 - (4 - 3)}{h}$$

= 5

$$L.f'(1) = \lim_{h \rightarrow 0^-} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{4(1+h)^2 - 3(1) - (4 \cdot 1^2 - 3 \cdot 1)}{h}$$

$$= \lim_{h \rightarrow 0^-} \frac{4(1+h)^2 - 3 - (4 - 3)}{h}$$

= 5

$$(2) \text{ A fn } f(x) = \begin{cases} 1 & ; x < 0 \\ 1 + \sin x & ; 0 \leq x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2 & ; x \geq \frac{\pi}{2} \end{cases}$$

Investigate the continuity and differentiability of the fn $f(x)$ at $x = \frac{\pi}{2}$

so/:- When $x > \frac{\pi}{2}$ then $f(x) = 2 + (x - \frac{\pi}{2})^2$

Indeterminate form.

$\ln 0 = -\infty$, $\ln \infty = \infty$, $\ln 1 = 0$, $\ln e = 1$, $e^\infty = \infty$, $e^{-\infty} = 0$
 (i) ∞^0 , $\frac{0}{0}$, $\frac{\infty}{\infty}$, $\infty \times \infty$, $\frac{a}{0}$, $\frac{1}{0}$, $\infty - \infty$, $0 + \infty$

$\infty \times 0$, 0^0 , $0 \times \infty$,

(ii) Note: $\frac{\infty}{\infty} = 1$, $\frac{1}{\infty} = 0$, $\frac{0}{\infty} = 0$.

Then may be applicable of L Hospital's theorem

(iii) $\frac{1}{(1+x)^{-1}} = 1 + x + x^2 + x^3 + \dots$
Important Expansion Series.

(iv) $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$

(v) $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

(vi) $e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$

(vii) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

(viii) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$

(ix) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$

(x) $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$

(xi) $\tan x = x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots$

(xii) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$

* (i) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ (ii) $\lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$ (iii) $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

(iv) $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$ (v) $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$

(vi) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$

$$1. \lim_{x \rightarrow 0} \frac{x e^x - \log(1+x)}{x^2} \left[\frac{0}{0} \therefore \text{using L'Hopital's rule} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{(e^x + x e^x) - \frac{1}{1+x}}{2x} \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{e^x - (e^x + x e^x) + \frac{1}{(1+x)^2}}{2}$$

$$\Rightarrow \frac{1 + 1 + 0 + \frac{1}{(1+0)^2}}{2}$$

$$\Rightarrow \frac{3}{2} \text{ Ans.}$$

$$2. \lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x - 4x}{x^5} \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{e^x + e^{-x} + 2 \cos x - 4}{5x^4} \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \sin x}{20x^3} \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2 \cos x}{60x^2} \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{e^x - e^{-x} + 2 \sin x}{120x} \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{e^x + e^{-x} + 2 \cos x}{120}$$

$$= \frac{1 + 1 + 2 \cdot 1}{120} = \frac{4}{120} = \frac{1}{30}$$

$$\frac{3}{2} \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} - \cos \theta - \sin \theta}{(4\theta - \pi)^2}$$

$$\Rightarrow \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\sin \theta - \cos \theta}{2(4\theta - \pi) \cdot 4} \quad \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos \theta + \sin \theta}{8 \cdot 4}$$

$$\Rightarrow \frac{\cos \frac{\pi}{4} + \sin \frac{\pi}{4}}{\frac{32}{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}}$$

$$= \frac{\frac{1}{\sqrt{2}}}{32}$$

$$= \frac{\frac{1}{\sqrt{2}}}{32}$$

$$= \frac{1}{16\sqrt{2}} \text{ Ans.}$$

$$\textcircled{4} \quad \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} \quad \left[\frac{0}{0} \right]$$

$$\Rightarrow \lim_{x \rightarrow a} \frac{n x^{n-1}}{1}$$

$$\Rightarrow n a^{n-1}$$

$$\textcircled{5} \lim_{x \rightarrow 0} \frac{x - \sin x}{\tan^3 x}$$

$$\times \lim_{x \rightarrow 0} \frac{x - (x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \infty)}{(\tan x)^3}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{x - x + \frac{x^3}{3!} - \frac{x^5}{5!} + \dots \infty}{(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \infty)^3} \\ & \lim_{x \rightarrow 0} \frac{\frac{x^3}{3!} + \frac{x^5}{5!} - \dots \infty}{x^3 (1 + \frac{x^2}{3} + \frac{2}{15}x^4 + \dots \infty)} \end{aligned}$$

$$\textcircled{6} \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

$$= \lim_{x \rightarrow 3} \frac{2x - 0}{1 - 0}$$

$$= 2 \times 3$$

$$\textcircled{7} \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{(\sqrt{1+x} - 1)(\sqrt{1+x} + 1)}{x(\sqrt{1+x} + 1)}$$

$$= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{1+x} + 1)}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x} + 1} = \frac{1}{\sqrt{1+0} + 1} = \frac{1}{2}$$

$$(7) \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x} = \frac{1}{2}$$

$$(8) \lim_{x \rightarrow \infty} \frac{x^{3/2} - x^{1/2}}{\sqrt{x} - \sqrt{x}} \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^{3/2} \cdot x^{1/2}}{\frac{1}{2} x^{-1/2}}$$

$$= x^2$$

$$= \frac{x^2}{\frac{1}{2} x^{-1/2}}$$

$$(9) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{0 + \sin x}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{2}$$

$$= \frac{1}{2}$$

$$\begin{aligned} & \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{1 - \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right)}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x^2}{2!} - \frac{x^4}{4!} + \dots}{x^2} \\ &= \frac{1}{2!} - 0 \\ &= \frac{1}{2} \end{aligned}$$

$$(10) \lim_{x \rightarrow 0} x^2 \log x \quad (0 \times \infty)$$

$$= \lim_{x \rightarrow 0} \frac{\log x}{x^{-2}}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-2x^{-3}}$$

$$= \lim_{x \rightarrow 0} \frac{1}{-2x^2}$$

$$= \lim_{x \rightarrow 0} \left(-\frac{1}{2x^2}\right)$$

$$= \lim_{x \rightarrow 0} (-\infty) = -\infty$$

$$(1) \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$$

$$2 \lim_{x \rightarrow 0} \frac{\sec^3 x - \cos x}{x^3}$$

$$3 \lim_{x \rightarrow 0} \frac{2 \sec^3 x \tan x + \sin x}{6x}$$

$$2 \lim_{x \rightarrow 0} \frac{2[\sec^4 x + \tan^2 x \cdot 2 \sec^2 x] + \cos x}{6}$$

$$2 \frac{2[1+0] + 1}{6}$$

$$2 \frac{2+1}{6}$$

$$2 \frac{3}{6}$$

= 1

Differentiation

$$y = f(x) \quad (I)$$

$$y + \Delta y = f(x + \Delta x) \quad (II)$$

$$\Delta y = f(x + \Delta x) - f(x)$$

$$\Rightarrow \frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$\text{or } \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

$$= \frac{dy}{dx}, \frac{d}{dx} f(x), f'(x)$$

or $\frac{dy}{dx}$ or $\frac{d}{dx} f(x)$ or $f'(x)$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow x = a \rightarrow$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Differential coefficient

① By defⁿ, find the differential coeff of x^n .

Solⁿ:- let, $f(x) = x^n = y$

$$\Rightarrow f(x+h) = (x+h)^n$$

By defⁿ, we know,

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[x^n \left(1 + \frac{h}{x} \right)^n - x^n \right]$$

$$= \lim_{h \rightarrow 0} \frac{x^n}{h} \left[\left(1 + \frac{h}{x} \right)^n - 1 \right]$$

$$= x^n \lim_{h \rightarrow 0} \frac{1}{h} \left[x^n + n \frac{h}{x} + \frac{n(n-1)}{2!} \frac{h^2}{x^2} + \dots \right]$$

$$= x^n \lim_{h \rightarrow 0} \frac{h}{h} \left[\frac{n}{x} + \frac{n(n-1)}{2!} \frac{h}{x^2} + \dots \right]$$

$$\frac{d}{dx} (x^n) = x^{n-1}$$

$$= nx^{n-1}$$

$$\text{If } e^x$$

$$\text{let } y = f(x) = e^x$$

$$f(x+h) = e^{x+h}$$

$$\text{By def } \frac{d}{dx} y = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\frac{d}{dx} e^x$$

$$= \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{e^x (e^h - 1)}{h}$$

$$\frac{d}{dx} e^x = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$$

$$= e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

$$= e^x \lim_{h \rightarrow 0} \frac{1}{h} \left[1 + \frac{h}{1!} + \frac{h^2}{2!} + \dots \right]$$

$$= e^x \lim_{h \rightarrow 0} \left[1 + \frac{h}{1!} + \frac{h^2}{2!} + \dots \right]$$

$$= e^x \cdot 1 = e^x$$

A.W $e^x, \ln(x), \sin x, \cos x, \tan x, \cot x, \sec x, \csc x$

$\log_a x, \tan^{-1} x, \sin^{-1} x, \cot^{-1} x, \sec^{-1} x, \csc^{-1} x$

$$\sqrt{x}, \left[\log_a x = \frac{1}{x} \log_e x \right] = \ln x \log_e e$$

(Chapter - two) (3)
ordinary differentiation

Differential Coefficients

Mathematics

1. principal Methods:-

General defⁿ of d.c.

$$y = f(x) \text{ --- (I)}$$

$$y + \delta y = f(x + \delta x) \text{ --- (II)}$$

$$\{ (II) - (I) \} \rightarrow$$

$$\delta y = f(x + \delta x) - f(x)$$

$$\Rightarrow \frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x}$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

$$\text{or, } \frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

$$(5) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$x = a \text{ ---}$$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

$$\frac{dy}{dx}$$

$$\frac{d}{dx}$$

$$f$$

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 কোয়ার্টার সামনে, কলেজ রোড, চকবাজার, টাউগ্রাম।
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 মোবাইল : 01775943269

Find $\frac{dy}{dx}$ for the following function by using Principal Method. or by defn, find the d. of

1. $y = f(x) = x^n, e^x, a^x, \ln x$ or $\log_e x, \log_a x$

2. $y = f(x) = \sin x, \cos x, \tan x, \csc x, \sec x, \cot x$

3. $y = \sin^{-1} x, \cos^{-1} x, \tan^{-1} x, \csc^{-1} x, \sec^{-1} x, \cot^{-1} x$

M RK $\rightarrow e^{\sqrt{x}}, x^{\sqrt{x}}, \cos x, \sin x, e^{5x+9}$!

Derive first derivatives from the following functions.

(1) ~~$y = \sec \tan^{-1} x$~~ From H.S.C BOOK.
or,

(1) $y = \sec \tan^{-1} x$ (2) $y = \ln(x^2+2)$ (3) $y = 5x^{3/5} + \tan^{-1} 2x$

(4) $\log_a x + \log_x a$ (5) $y = (\ln \sin x)^2$ (6) $y = \frac{x^n}{\log_a x}$

(7) $y = \tan^{-1} \frac{a+bx}{b-ax}$ (8) $y = \sqrt[3]{x^2 + \sqrt{x}}$

(9) $y = 10^{\ln \sin x}$ (10) $y = e^{\sin x} - (\sin x)^e$

(11) $y = x^{x^2}$ (12) $y = x e^x$ (13) $y = (\sin x)^{\ln x}$

(14) $y = \sin^{-1}(\ln x^2)$ (15) $y = (\sec \tan^{-1} x) (x^5 + x^{1/3})^{2/3}$

(16) $y = (\ln \sin x)^2 \{ \log_a x + \log_x a \}$ (17) $y = \frac{x^n (x^5 + x^{1/3})^{3/5}}{\log_a x}$

(18) $y = \frac{\ln(x^2+2) (x^5 + x^{1/3})}{x^2 e^x}$ (19) $y = \frac{x^n (x^5 + x^{1/3})^{3/5}}{x^2 e^x}$

Successive differentiation.

Find n^{th} derivative for the following f's:

(i) $y = \ln x$

(ii) $y = x^m$ and x^{-m}

(iii) $y = a^x$

(iv) $y = e^{ax}$

(v) $y = \sin x$ & $\cos x$.

(vi) $y = \ln(ax+b)$

(vii) $\sin(ax+b)$ and $\cos(ax+b)$

MRK

(i) $x^m e^{ax}$

(ii) $y = e^x (ax)$

State and prove Leibnitz's theorem:-

Prove the following:

(1) If $y = \tan^{-1} x$ show that $(1+x^2) y_{n+1} + 2nx y_n + x^2 y_{n-1} - n y_{n-1} = 0$

(2) If $y = e^{ax} \cos^{-1} x$ then show that $(1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2+a^2) y_n = 0$

(3) If $y = e^{ax} \sin^{-1} x$ prove that $(1-x^2) y_{n+2} - (2n+1)x y_{n+1} - (n^2+a^2) y_n = 0$

(4) If $x = \sin\left(\frac{1}{m} \log\right)$ then $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2 + m^2)y_n = 0$

or
If $x = \left(\frac{1}{a} \log\right)$ then prove that $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2 + a^2)y_n = 0$.

(5) Define if $\log x = \tan^{-1} u$, then show that
 $(1+x^2)y_{n+2} + (2nx + 2x - 1)y_{n+1} + n(n+1)y_n = 0$.

Leibnitz's theorem: Statement:- If u & v are fns of x then the n th derivative of the product of u & v is

$$(uv)_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + n C_r u_{n-r} v_r + \dots$$

where the suffixes of u & v indicate the order of diff. w.r. to x .

Proof:- Let $y = uv$, integrating w.r. to x

$$y_1 = u_1 v + u v_1$$

$$y_2 = u_2 v + u_1 v_1 + u_1 v_1 + u_2 v = u_2 v + 2u_1 v_1 + u_2 v$$

$$y_3 = u_3 v + u_1 v_2 + 2(u_1 v_2 + u_2 v_1) + u_2 v_1 + u_3 v$$

$$= u_3 v + 3u_1 v_2 + 3u_2 v_1 + u_3 v$$

$$y_3 = u_3 v + 3 C_1 u_1 v_{3-2} + 3 C_2 u_2 v_{3-1} + u_3 v \quad \left[\begin{matrix} 3 C_1 = 3 \\ 3 C_2 = 3 \end{matrix} \right]$$

Since the theorem is true for $n=2$ & $n=3$

then it is true for any n Hence we can say

$$(uv)_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + n C_r u_{n-r} v_r + \dots$$

where $y_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + n C_r u_{n-r} v_r + \dots$

Successive Differentiation

If y is a function of x , then the differential coefficient $\frac{dy}{dx}$ is called first differential coefficient of y .

In short, If $y = f(x)$, the successive derivatives are also denoted as follows:

#	First time	2nd time	Third time	...	n times
	$\frac{dy}{dx}$	$\frac{d^2y}{dx^2}$	$\frac{d^3y}{dx^3}$	...	$\frac{d^ny}{dx^n}$
	y_1	y_2	y_3	...	y_n
	y'	y''	y'''	...	$y^{(n)}$
	$\frac{d}{dx} \rightarrow D y$	$D^2 y$	$D^3 y$	...	$D^n y$
	$f'(x)$	$f''(x)$	$f'''(x)$	...	$f^{(n)}(x)$

The process of Differentiating the function again and again is called successive differentiation.

* Obtain n^{th} derivatives of following:

(1) $y = \frac{1}{x}$

(2) $y = \ln x$

(3) $y = x^n$

(4) $y = \cos x$

(5) $y = \sin x$

(6) $y = a^x$

(7) $y = e^{ax} \text{ (or) } e^{ax+b}$

(8) $\sin(ax+b)$ and $\cos(ax+b)$

(9) $\ln(ax+b)$

(1) Given that

we show $D^n y = n!$

$$y = x^n$$

Diff. w.r. to x

$$D^1 y = n x^{n-1}$$

$$D^2 y = n(n-1) x^{n-2}$$

$$D^3 y = n(n-1)(n-2) x^{n-3}$$

$$D^4 y = n(n-1)(n-2)(n-3) x^{n-4}$$

$$D^n y = n(n-1)(n-2)(n-3) \dots (n-4) \dots 4 \cdot 3 \cdot 2 \cdot 1 x^{n-n}$$

$$= n!$$

$$= n!$$

$$= n!$$

$$D^n y = n!$$

(2) $y = \ln x$

Diff w.r. to x

$$y_1 = \frac{1}{x} = x^{-1}$$

$$y_2 = (-1) x^{-2}$$

$$y_3 = 2 x^{-3} = (-1)^2 x^{-3}$$

$$y_4 = (-1)^3 (-3) x^{-4} = (-1)^3 x^{-4}$$

$$y_n = (-1)^{n-1} (n-1) x^{-n}$$

1. If $y = \frac{1}{x}$ find $\frac{dy}{dx}$

Soln. Given that $y = \frac{1}{x}$
or $y = x^{-1}$

Diff. w.r. to x

$$y_1 = (-1) x^{-2}$$

Again

$$y_2 = (-1)(-2) x^{-3}$$

$$= (-1)^2 2 x^{-3}$$

$$y_3 = (-1)^2 (-3) 2 x^{-4}$$

$$= (-1)^3 2^2 x^{-4}$$

$$y_n = (-1)^n L_n x^{-(n+1)}$$

$$y_{n+1} = (-1)^{n+1} L_{n+1} x^{-(n+2)}$$

(iv) $y = \cos x$

$$y_1 = -\sin x$$

$$= \cos\left(\frac{\pi}{2} + x\right) \left[\cos(90^\circ + \theta) = -\sin \theta \right]$$

$$y_2 = -\sin\left(\frac{\pi}{2} + x\right)$$

$$= \cos\left\{ \frac{\pi}{2} + \left(\frac{\pi}{2} + x \right) \right\}$$

$$y_2 = \cos\left(\frac{\pi}{2} + x\right)$$

$\frac{dy}{dx}$
1
2
3
4
5
6
7
8
9
10
11
12

$$= a \cos\left[\frac{\pi}{2} + (ax+b)\right]$$

$$\text{Again, } y_2 = a^2 \left[-\sin\left(\frac{\pi}{2} + (ax+b)\right) \right] = a^2 \cos\left(\frac{\pi}{2} + (ax+b)\right)$$

$$y_3 = a^3 \left[-\sin\left\{2\frac{\pi}{2} + (ax+b)\right\} \right]$$

$$= a^3 \cos\left[\frac{\pi}{2} + 2\cdot\frac{\pi}{2} + (ax+b)\right]$$

$$= a^3 \cos\left[3\frac{\pi}{2} + (ax+b)\right]$$

$$y_n = a^n \cos\left[n\frac{\pi}{2} + (ax+b)\right]$$

⑥ If $y = \ln(ax+b)$ show that $y_n = \frac{a^n (-1)^{n-1} (n-1)!}{(ax+b)^n}$ when $n \in \mathbb{N}$
diff w.r. to x

$$y_1 = \frac{a}{ax+b} = a \cdot (ax+b)^{-1}$$

$$y_2 = a^2 (-1) (ax+b)^{-2}$$

$$y_3 = a^3 a (-1) \cdot (-2) \cdot 1 (ax+b)^{-3}$$

$$= a^3 (-1)^2 2! (ax+b)^{-3}$$

$$y_4 = a^4 (-1)^3 3! (ax+b)^{-4}$$

$$y_4 = a^4 (-1)^3 3! (ax+b)^{-4}$$

$$= a^4 (-1)^3 3! (ax+b)^{-4}$$

$$y_n = \frac{a^n (-1)^{n-1} (n-1)!}{(ax+b)^n}$$

More problems from H.S.C book.

Leibniz Theorem.

Verified with
Lab
(11/10/2019)

State and prove Leibniz Theorem.

State statement: If u and v are functions of x , then the n th derivative of the product of u and v is

$$(uv)_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + n C_r u_r v_{n-r} + \dots + u v_n$$

where the suffixes of u and v indicate the order of differentiation w.r. to x .

Proof:- let $y = uv$

$$\therefore y_1 = u_1 v + v u_1 = u_1 v + u v_1$$

Again differentiate

$$y_2 = u_1 v_1 + u v_2 + v_1 u_1 + v u_2$$

$$= u_2 v + 2 u_1 v_1 + u v_2$$

Again differentiate,

$$y_3 = u_2 v_1 + v u_3 + 2(u_1 v_2 + v_1 u_2) + u v_3 + v_2 u_2$$

$$= u_3 v + 3 u_2 v_1 + 2 u_1 v_2 + u v_3$$

Since the theorem is true for $n=2$ and $n=3$ then it is true for any n . Hence

we can say

$$(uv)_n = u_n v + n C_1 u_{n-1} v_1 + n C_2 u_{n-2} v_2 + \dots + n C_r u_r v_{n-r} + \dots + u v_n$$

Problem 1:

If $y = \tan^{-1} x$ show that $(1+x^2) \frac{d^n y}{dx^n} + n(1+x^2) \frac{d^{n-1} y}{dx^{n-1}} = 0$

Soln:-

Given, $y = \tan^{-1} x$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2) \frac{dy}{dx} = 1$$

Differentiating both sides w.r.t. x by Leibniz theorem we have,

$$(1+x^2) \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 2x \frac{dy}{dx} = 0$$

$$\Rightarrow (1+x^2) \frac{d^2 y}{dx^2} + 2nx \frac{d^{n-1} y}{dx^{n-1}} + \frac{n(1+x^2)}{1+x^2} \frac{d^{n-1} y}{dx^{n-1}} = 0$$

$$\Rightarrow (1+x^2) \frac{d^2 y}{dx^2} + 2nx \frac{d^{n-1} y}{dx^{n-1}} + n \frac{d^{n-1} y}{dx^{n-1}} - n \frac{d^{n-1} y}{dx^{n-1}} = 0$$

(Hence proved)

Problem 2:

If $y = e^{a \cot^{-1} x}$ then show that

$$(1+x^2) \frac{d^2 y}{dx^2} - (2x+1) \frac{dy}{dx} - (a^2+x^2) y = 0$$

Soln:-

Given, $y = e^{a \cot^{-1} x}$

$$\Rightarrow \log y = \log e^{a \cot^{-1} x} \quad \bigg| \quad \log e^x = x$$

$$\Rightarrow \log y = a \cot^{-1} x$$

Diff. w.r.t. x we get,

$$\frac{1}{\sqrt{1-x^2}} = \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow (\sqrt{1-x^2}) y_1 = -x y_0$$

$$\Rightarrow (1-x^2) y_1' = x y_0'$$

$$\Rightarrow (1-x^2) 2y_1 y_2 + y_1' (-2x) = x 2y_1 y_2'$$

$$\Rightarrow (1-x^2) y_2 - 2xy_1 - x^2 y_1' = 0 \quad [\text{dividing both sides by } 2y_1]$$

$$\Rightarrow (1-x^2) y_2 - 2xy_1 - x^2 y_1' = 0 \quad \text{--- (1)}$$

Differentiating both sides n times by Leibniz theorem we get.

$$(1-x^2) y_{n+2} + n y_{n+1} (-2x) + \frac{n(n-1)}{2} y_n (-2) - x^2 y_{n+1}' = 0$$

$$- n y_n - x^2 y_n' = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - 2nx y_{n+1} - n^2 y_n + 2xy_n - x^2 y_{n+1}' - n y_n - x^2 y_n' = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + x^2) y_n = 0$$

(Hence proved)

H.W:-

3

Let $y = e^{ax} \sin bx$. Prove that

$$(1-a^2) y_{n+2} - (2n+1) x y_{n+1} - (n^2 + x^2) y_n = 0$$

If $y = a \sin^{-1} x + b \cos^{-1} x$, show that

$$(1-x^2)y'' - (2nx+1)y' - n^2 y = 0$$

Soln:- Given that,

$$y = a \sin^{-1} x + b \cos^{-1} x \quad (1)$$

Diff (1) w.r. to x

$$y' = \frac{a}{\sqrt{1-x^2}} + \frac{b}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} y' = a + b$$

$$\Rightarrow (1-x^2) y' = (a+b)x$$

$$\Rightarrow (1-x^2) 2y' y'' = (a+b) 2x y'$$

$$\Rightarrow (1-x^2) y'' = (a+b) y' \quad \left[\text{Dividing both sides by } 2y' \right]$$

$$(1-x^2) y'' + n_1 (-2x) y' + n_2 (-2) y = 0$$

$$\Rightarrow (1-x^2) y'' - 2nx y' - n^2 y = 0$$

$$\Rightarrow (1-x^2) y'' - (2n+1) x y' - n^2 y = 0$$

Problem :

If $x = \sin\left(\frac{1}{m} \ln y\right)$ then

$$(1-x^m) y_{n+2} - (2nx) y_{n+1} - (n^2 x^m) y_n = 0$$

Solⁿ:- Given $x = \sin\left(\frac{1}{m} \ln y\right)$

$$\Rightarrow \sin^{-1} x = \frac{1}{m} \ln y$$

$$\Rightarrow m \sin^{-1} x = \ln y$$

$$\Rightarrow y = e^{m \sin^{-1} x} \quad \text{--- (1)}$$

Diff. (1) w.r. to x

$$y_1 = e^{m \sin^{-1} x} \cdot \frac{m}{\sqrt{1-x^2}} \quad \text{--- (2)}$$

$$\Rightarrow \sqrt{1-x^2} y_1 = m y$$

$$\Rightarrow (1-x^2) y_1^2 = m^2 y^2$$

$$\Rightarrow (1-x^2) 2 y_1 y_2 - 2 x y_1^2 = m^2 2 y y_1$$

$$\Rightarrow (1-x^2) y_2 - x y_1^2 = m^2 y y_1$$

Leibniz theorem,

$$\Rightarrow (1-x^2) y_{n+2} + n C_1 (-2x) y_{n+1} + n C_2 (-2) y_n$$

$$- \{ x y_{n+1} + n C_1 (1) y_n \} - m^2 y_n = 0$$

$$\Rightarrow (1-x^2) y_{n+2} - 2nx y_{n+1} - \frac{n(n-1)}{2!} 2x^n - n y_{n+1} - m^2 y_n = 0$$

$$\Rightarrow (1-\lambda^2)y_{n+2} - (2n+1)\lambda y_{n+1} - (n^2-n+1+m)y_n = 0$$

$$\Rightarrow (1-\lambda^2)y_{n+2} - (2n+1)\lambda y_{n+1} - (n^2+m)y_n = 0$$

* If $y = f(x) = \sin^{-1} x$ then show that

(i) $(1-\lambda^2)y_2 - \lambda y_1 = 0$

(ii) $(1-\lambda^2)y_{n+2} - (2n+1)\lambda y_{n+1} - n^2 y_n = 0$

(iii) $(y_n)_0 = (n-2)^2 (n-4)^2 \dots 3^2 \cdot 1^2$ for n is odd numbers.

(iv) $(y_n)_0 = 0$ for n is even number.

Soln:- Given, $y = \sin^{-1} x$ --- (i)

$$\Rightarrow y_1 = \frac{1}{\sqrt{1-x^2}} \quad \text{--- (ii)}$$

$$\Rightarrow y_1 (\sqrt{1-x^2}) = 1$$

$$\Rightarrow y_1^2 (1-x^2) = 1$$

$$\Rightarrow (1-x^2) 2x y_2 - 2x y_1^2 = 0$$

$$\Rightarrow (1-x^2) y_2 - x y_1 = 0 \quad \text{--- (iii)}$$

Applying Leibnitz's theorem.

$$(1-x^2)y_{n+2} + n y_{n+1} (-2x) + \frac{n(n-1)}{2!} y_n (x^2) - \lambda y_{n+1} = 0$$

$$\Rightarrow (1-x^2)y_{n+2} - 2\lambda y_{n+1} - n(n-1)y_n - \lambda y_{n+1} = 0$$

State
Stations

Chapter 2

(1691)

Stressment:

differentiable if $f'(x)$ exists

differentiable if $f'(x)$ exists

Interval $a < x < b$.

$$\alpha \text{ and } (111) \quad f(\alpha) = f(\beta)$$

then there exists at least one value of x

(not c) between $\eta_i^{\wedge}$ and b [$i, a \leq c \leq b$], such that

11-1

Proof: As the function $f(x)$ is continuous on the interval $a \leq x \leq b$, and $f(a) = f(c)$, $f(x)$ has at least a maximum or a minimum or both in the interval. Either $f(x)$ has maximum value N at $x = c$ i.e., $f(c) = N$ or a minimum value m at $x = c$ i.e., $f(c) = m$, (Hing in the interval (a, b)).

Answer for Ques:-

When from the Concord to the

normal E_n^1 for $M = m = f(a) = f(b) = f(c)$

work done. $\int_{C(0)}^{\infty} \omega_{\text{Klein}}$

Integral theorem and Expansions

(Chapter - 3)

(4)

State and prove Rolle's Theorem. (1691)

Statement:- If (i) $f(x)$ is continuous in the close interval $a \leq x \leq b$

(ii) $f(x)$ is differentiable i.e. $f'(x)$ exists in the open interval $a < x < b$.

and (iii) $f(a) = f(b)$

then there exists at least one value of x (say c) between a and b [i.e. $a < c < b$], such that $f'(c) = 0$

Proof:- As the function $f(x)$ is continuous in the interval $a \leq x \leq b$, and $f(a) = f(b)$; $f(x)$ has at least a maximum or a minimum or both in the interval. Either $f(x)$ has maximum value M at $x = c$ i.e. $f(c) = M$ or a minimum value m at $x = c$ i.e. $f(c) = m$, lying in the interval (a, b) .

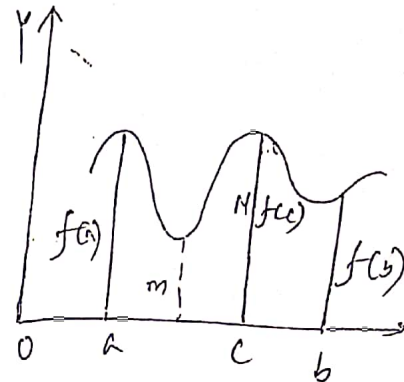
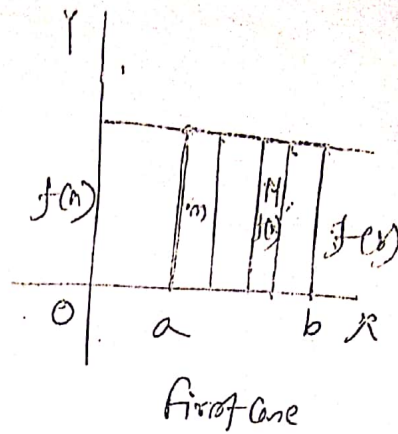
There are two cases:- (i) $M = m$

When $M = m$, then $f(x)$ be constant in the interval $[a, b]$ i.e. $M = m = f(x) = f(a) = f(b)$

and hence $f'(c) = 0$ when $a < c < b$

2nd Case

When $M \neq m$, then at least one value of $f(x)$ is different from $f(c)$ and $f(a)$ in the interval. Let $f(c) = M$ be different from them, where c lies within the interval. Since $f(c)$ is the maximum value of the function $f(x)$, then $f(c+h) - f(c) \leq 0$.



whether h is $+$ or $-$.

Thus we have,

$$\frac{f(c+h) - f(c)}{h} \leq 0 \text{ when } h > 0 \text{ and } \frac{f(c+h) - f(c)}{h} \geq 0, \text{ when } h < 0$$

But from the statement of the theorem we see that $f(x)$ exists at $x = c$.

$$\text{Now, } \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} \leq 0 \text{ and}$$

$$\lim_{h \rightarrow 0^-} \frac{f(c+h) - f(c)}{h} \geq 0.$$

$$\Rightarrow f'(c) \geq 0 \text{ --- (1)}$$

$$\& f'(c) \leq 0 \text{ --- (2)}$$

Problem-1: Verify the graph of Rolle's theorem for the function $f(x) = x^2 - 3x + 2$ in the interval $(1, 2)$

Soln: When $x = 1$, then $f(1) = 1 - 3 + 2 = 0$.

When $x = 2$, then $f(2) = 4 - 6 + 2 = 0$ //

$$\therefore f(1) = f(2)$$

$$\text{Now, } f'(x) = 2x - 3$$

$$\text{If } f'(x) = 0 \text{ then } 2x - 3 = 0$$

$$\Rightarrow x = \frac{3}{2} = 1.5$$

Thus we see that $f(x)$ is continuous in $1 \leq x \leq 2$; $f'(x)$ exists in $1 < x < 2$ and $f(1) = f(2)$. There exists a point $x = 1.5$ within the interval $(1, 2)$ such that $1 < 1.5 < 2$ where $f'(x) = 0$ i.e., $f'(1.5) = 0$.

Problem-2: $f(x) = (x-2)(x-3)(x-4)$ in $2 \leq x \leq 3$

Soln: Given, $f(x) = (x-2)(x-3)(x-4)$; $2 \leq x \leq 3$.

$$\text{or, } f(x) = x^3 - 9x^2 + 26x - 24$$

$$\text{or, } f'(x) = 3x^2 - 18x + 26$$

$f(2) = f(3)$

$f(2) = f(3)$

Hence Rolle's theorem condition is satisfied.
Rolle's theorem says there exists at least
one point 'c' in $(2, 3)$, such that $f'(c) = 0$.

$$\text{Now, } 2c^2 - 18c + 26 = 0.$$

$$\Rightarrow c = \frac{18 \pm \sqrt{18^2 - 4 \cdot 2 \cdot 26}}{2 \cdot 2}$$

$$= \frac{18 \pm \sqrt{324 - 208}}{4} = \frac{18 \pm \sqrt{116}}{4}$$

$$= \frac{18 \pm \sqrt{4 \cdot 29}}{4} = 3 \pm \frac{\sqrt{29}}{2}$$

$$= 3 \pm \frac{1}{2}\sqrt{29}$$

$$= 3.52, 2.43$$

Clearly, $2.43 \in (2, 3)$.

Hence Rolle's theorem is verified.

Problem-3: verify Rolle's theorem

$$f(x) = 2x^3 + x^2 - 4x - 2.$$

Soln:- Given, $f(x) = 2x^3 + x^2 - 4x - 2$

Since, $f(x)$ is a polynomial in x ,

so $f(x)$ is continuous and differentiable.

$$2x^3 + x^2 - 4x - 2 = 0$$

$$1) \quad 2x^3 + x^2 - 4x - 2 = 0$$

$$2) \quad x^2(2x+1) - 2(2x+1) = 0 \quad (2x+1)$$

$$\Rightarrow (2x+1)(x^2-2) = 0$$

$$\therefore x = -\sqrt{2}, -\frac{1}{2}, \sqrt{2}$$

$$\therefore, f(-\sqrt{2}) = (-\sqrt{2})^3 - 2(-\sqrt{2}) = 0$$

Hence $f(x)$ satisfies the condition of Rolle's theorem. So there exists two points c_1 & c_2 s.t. $f'(c_1) = 0$ & $f'(c_2) = 0$.

$$3) \quad f'(x) = 6x^2 + 2x - 4 = 0$$

$$\Rightarrow 6x^2 + 6x - 4x - 4 = 0$$

$$\Rightarrow 6x(x+1) - 4(x+1) = 0$$

$$\Rightarrow x = -1, \quad x = \frac{4}{6} = \frac{2}{3}$$

$$\text{clearly, } c_1 = -1 \in (-\sqrt{2}, -\frac{1}{2}) \&$$

$$c_2 = \frac{2}{3} \in (-\frac{1}{2}, \sqrt{2})$$

Hence Rolle's theorem is verified.

Statement:- Let f be a function defined on a closed interval $a \leq x \leq b$. $f'(x)$ exists in the open interval $a < x < b$, then exists at least one point c , $a < c < b$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

Proof:- Let $\phi(x) = f(x) - Ax$ — (i)
 Let $\phi(x)$ be an auxiliary fn, where A is constant, so chosen that
 $\phi(b) = \phi(a)$

$$\text{i.e., } f(b) - Ab = f(a) - Aa$$

$$\Rightarrow f(b) - f(a) = Ab - Aa$$

$$\Rightarrow A = \frac{f(b) - f(a)}{b - a} \quad \text{--- (ii)}$$

putting the value of A in eqn (i)

$$\phi(x) = f(x) - \left\{ \frac{f(b) - f(a)}{b - a} \right\} x \quad \text{--- (iii)}$$

Differentiating eqn (iii) w.r. to x .

$$\phi'(x) = f'(x) - \frac{f(b) - f(a)}{b - a} \quad \text{--- (iv)}$$

With the help of eqn (ii) and the given conditions we find that $\phi'(c)$ is continuous in $a < c < b$ and differentiable in

∴ Hence $\phi'(c) = 0$ where $c \in (a, b)$.
---(iv)

From (iv) $\phi'(x) = f'(x) - A$

$$\Rightarrow \phi'(c) = f'(c) - A$$

$$f'(c) - A = 0$$

$$\Rightarrow f'(c) = A$$

$$= \frac{f(b) - f(a)}{b - a}$$

$$\therefore f(b) - f(a) = f'(c) (b - a)$$

Which is called first mean value theorem of Differential Calculus and also known as Lagrange's form of the mean value theorem.

problem:- verify mean value theorem for the

$$f(x) = 3 + 2x - x^2 \text{ in } (0, 1)$$

Soln:- The given function, $f(x) = 3 + 2x - x^2$

Since $f(x)$ is polynomial, so $f(x)$ is continuous and differentiable in everywhere.

Let $f(x)$ be a function defined on the interval $[a, b]$.
differentiable on (a, b) . The MVT states that
if the mean value theorem is satisfied, then
there exists a point c in (a, b) such that

$$\text{But } f(b) = f(a) = 3.$$

$$\text{and } f(b) = f(a) = 3 + 2 \cdot 1 - 1 = 0$$

$$f'(x) = 2 - 2x$$

$$\therefore f'(c) = 2 - 2c$$

So, we find by mean value theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\Rightarrow 2 - 2c = \frac{4 - 3}{1 - 0} = \frac{1}{1} = 1$$

$$\Rightarrow 2 - 1 = 2c$$

$$c = \frac{1}{2} \in (0, 1)$$

Hence, the mean value theorem is
valid for given $f(x)$.

Q.10) Verify Rolle's theorem for the function
 $f(x) = 3x^3 + 7x^2 - 11x - 15$

Soln:- Given $f(x) = 3x^3 + 7x^2 - 11x - 15$

Since, $f(x)$ is a polynomial in x so $f(x)$ is continuous and differentiable in the $(-\infty, \infty)$

$$\text{Here } f(x) = 3x^3 + 7x^2 - 11x - 15 = 0$$

$$\Rightarrow 3x^3 + 3x^2 + 4x^2 + 4x - 15x - 15 = 0$$

$$\Rightarrow 3x^2(x+1) + 4x(x+1) - 15(x+1) = 0$$

$$\Rightarrow (x+1)(3x^2 + 4x - 15) = 0$$

$$\Rightarrow (x+1)(3x^2 + 9x - 5x - 15) = 0$$

$$\Rightarrow (x+1)\{3x(x+3) - 5(x+3)\} = 0$$

$$\Rightarrow (x+1)(x+3)(3x-5) = 0$$

$$\Rightarrow x = -1, x = -3, x = \frac{5}{3}$$

$$\Rightarrow x = -1, -3, \frac{5}{3}$$

$$\text{i.e., } f(-1) = f(-3) = f\left(\frac{5}{3}\right) = 0$$

Hence $f(x)$ satisfies the condition of Rolle's theorem. So there exist two points c_1 & c_2 s.t. $f'(c_1) = 0$ & $f'(c_2) = 0$.

11/11/1912

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Maclaurin's Theorem
(only for infinite and differentiable function)

Hypothesis:- Let $f(x)$ be a function of x which can be expanded in power of x then

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

Proof:- let $f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + \dots$ ①

Now successive differentiation w.r. to x we get, in eqn ①

$$f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \dots$$

$$f''(x) = 2 \cdot 1 a_2 + 2 \cdot 3 a_3 x + 3 \cdot 4 a_4 x^2 + \dots$$

$$f'''(x) = 1 \cdot 2 \cdot 3 a_3 + 2 \cdot 3 \cdot 4 a_4 x + \dots$$

$$\dots$$

Now, putting $x=0$ we get,

$$f(0) = a_0, \quad f'(0) = a_1, \quad f''(0) = 2! a_2, \quad f'''(0) = 3! a_3$$

$$\text{Similarly, } f^{(n)}(0) = n! a_n$$

putting all these values in eqn ① we get,

$$f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots + \frac{x^n}{n!} f^{(n)}(0) + \dots$$

$$f(x) = \tan x$$

$$f(0) = 0$$

$$f(x) = \sec x$$

$$f'(0) = 1$$

$$f''(0) = 2\sec x \cdot \sec x \tan x$$

$$= 2\sec^2 x \cdot \tan x \quad \therefore f''(0) = 0$$

$$f'''(x) = 2\sec^4 x + 4\sec^2 x \tan^2 x \quad \therefore f'''(0) = 2$$

By using Maclaurian's formula we get,

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$\therefore \tan x = 0 + x \cdot 1 + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!} \cdot 2 + \dots$$

$$\therefore \tan x = x + \frac{2x^3}{3!} + \dots$$

$$= x + \frac{2x^3}{3} + \dots$$

$$(3) \tan^{-1} x$$

$$f(x) = \tan^{-1} x$$

$$\therefore f(0) = \tan^{-1}(0) = 0$$

$$\therefore f'(0) = 1$$

$$f'(x) = \frac{1}{1+x^2}$$

$$= (1+x^2)^{-1} = 1 - x^2 + x^4 - x^6 + x^8 - \dots$$

$$f''(x) = -2x + 4x^3 - 6x^5 + \dots \quad \therefore f''(0) = 0$$

$$f'''(x) = -2 + 12x^2 - 30x^4 + \dots \quad \therefore f'''(0) = -2$$

$$f^{(4)}(x) = 24x - 120x^3 + \dots \quad \therefore f^{(4)}(0) = 0$$

$$f(x) = e^{mx}$$

$$f(x) = e^{mx} \quad ; \quad f(0) = e^0 = 1$$

$$f'(x) = me^{mx} \quad f'(0) = me^0 = m$$

$$f''(x) = m^2 e^{mx} \quad f''(0) = m^2 e^0 = m^2$$

$$f'''(x) = m^3 e^{mx} \quad f'''(0) = m^3 e^0 = m^3$$

$$e^{mx} = 1 + mx + \frac{m^2 x^2}{2!} + \frac{m^3 x^3}{3!} + \dots$$

$$\# \quad e^x \sin x$$

$$f(x) = e^x \sin x \quad \therefore f(0) = 0$$

$$f'(x) = e^x (\cos x + \sin x) \quad \therefore f'(0) = 1$$

$$f''(x) = e^x (-\sin x + \cos x) + (\cos x + \sin x) e^x$$

$$= -\sin x e^x + e^x \cos x + \cos x e^x + \sin x e^x$$

$$= 2e^x \cos x \quad f''(0) = 2$$

$$f'''(x) = -2e^x \sin x + 2e^x \cos x \quad f'''(0) = 2$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

Taylor's theorem

Statement:- Taylor series of degree n about a is

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2$$

$$+ \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

This formula $f(x)$ near a

* Find the third degree Taylor polynomial of $f(x) = \sqrt{x}$ with center $x_0 = 0$.

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